SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet

range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. Subnetting for University Network

Given Network: `192.168.20.0/24`

Lab	Subnet Mask	Prefix
Lab A	255.255.255.192	/26
Lab B	255.255.255.224	/27
Lab C	255.255.255.240	/28

Subnet allocation starts from the beginning of the network.

Step 1: Lab A (/26)

- Block Size = 64
- Network Address = 192.168.20.0
- Broadcast Address = 192.168.20.63

Host Range:

- First Host = 192.168.20.1
- Last Host = 192.168.20.62

Usable Hosts

2(32−26)−2=64−2=622^{(32-26)}-2=64-2=622(32−26)−2=64−2=62

Usable Hosts = 62

Step 2: Lab B (/27)

Next available address = 192.168.20.64

Block Size = 32

- **Network Address = 192.168.20.64**
- **Broadcast Address = 192.168.20.95**

Host Range

- **First Host = 192.168.20.65**
- **Last Host = 192.168.20.94**

Usable Hosts

25−2=32−2=302^{5}-2=32-2=3025−2=32−2=30

Usable Hosts = 30

Step 3: Lab C (/28)

Next available address = 192.168.20.96

Block Size = 16

- **Network Address = 192.168.20.96**
- **Broadcast Address = 192.168.20.111**

Host Range

- **First Host = 192.168.20.97**
- **Last Host = 192.168.20.110**

Usable Hosts

24−2=16−2=142^{4}-2=16-2=1424−2=16−2=14

Usable Hosts = 14

Final Answer

Lab	Network Address	Host Range	Broadcast	Usable Hosts
A (/26)	192.168.20.0	192.168.20.1 – 192.168.20.62	192.168.20.63	62
B (/27)	192.168.20.64	192.168.20.65 – 192.168.20.94	192.168.20.95	30
C (/28)	192.168.20.96	192.168.20.97 – 192.168.20.110	192.168.20.111	14

---

2. Subnet Analysis

Subnet Mask = 255.255.255.192 (/26)

Block Size

256−192=64256-192=64256−192=64

Subnet ranges are

- **0–63**
- **64–127**
- **128–191**
- **192–255**

System 1

IP = 192.168.1.130

130 lies between 128–191

Network Address

192.168.1.128

Broadcast Address

192.168.1.191

Host Range

192.168.1.129 – 192.168.1.190

Usable Hosts

64−2=6264-2=6264−2=62

✔ IP 192.168.1.130 belongs to this subnet.

System 2

IP = 192.168.1.190

190 also lies between 128–191

Network Address

192.168.1.128

Broadcast Address

192.168.1.191

Host Range

192.168.1.129 – 192.168.1.190

Usable Hosts

62

✔ IP 192.168.1.190 belongs to the same subnet.

Final Table

System	IP Address	Network Address	Host Range	Broadcast	In Range?	Usable Hosts
System 1	192.168.1.130	192.168.1.128	192.168.1.129–192.168.1.190	192.168.1.191	Yes	62
System 2	192.168.1.190	192.168.1.128	192.168.1.129–192.168.1.190	192.168.1.191	Yes	62

---

3. Organization Network

Given Network

172.16.0.0/16

Departments

Department	Prefix
HR	/25
Sales	/26
IT	/27
Admin	/28

HR (/25)

Network = 172.16.0.0

Broadcast = 172.16.0.127

Host Range

172.16.0.1 – 172.16.0.126

Usable Hosts

128−2=126128-2=126128−2=126

Sales (/26)

Next Network

172.16.0.128

Broadcast

172.16.0.191

Host Range

172.16.0.129 – 172.16.0.190

Usable Hosts

30? Wait /26 = 64 addresses

64−2=6264-2=6264−2=62

IT (/27)

Network

172.16.0.192

Broadcast

172.16.0.223

Host Range

172.16.0.193 – 172.16.0.222

Usable Hosts

30

Admin (/28)

Network

172.16.0.224

Broadcast

172.16.0.239

Host Range

172.16.0.225 – 172.16.0.238

Usable Hosts

14

Remaining Unused Range

Allocated until 172.16.0.239

Unused begins from

172.16.0.240

Remaining Network

172.16.0.240 – 172.16.255.255

Final Table

Department	Network	Host Range	Broadcast	Usable Hosts
HR (/25)	172.16.0.0	172.16.0.1 – 172.16.0.126	172.16.0.127	126
Sales (/26)	172.16.0.128	172.16.0.129 – 172.16.0.190	172.16.0.191	62
IT (/27)	172.16.0.192	172.16.0.193 – 172.16.0.222	172.16.0.223	30
Admin (/28)	172.16.0.224	172.16.0.225 – 172.16.0.238	172.16.0.239	14

---

4. Company Subnetting

Given

Network = 172.16.10.0/24

Subnet Mask = /27

Total Subnets

Borrowed bits

27−24=327-24=327−24=3 23=82^3=823=8

Total Subnets = 8

Hosts per Subnet

25−2=302^5-2=3025−2=30

Usable Hosts = 30

Find subnet of 172.16.10.78

Block Size

256−224=32256-224=32256−224=32

Subnet ranges

0–31

32–63

64–95

96–127

...

78 lies inside 64–95

Network Address

172.16.10.64

Broadcast

172.16.10.95

Host Range

172.16.10.65 – 172.16.10.94

Does 172.16.10.95 belong?

172.16.10.95 is the Broadcast Address

It is not usable as a host, although it belongs to the same subnet.

Final Answer

Item	Result
Total Subnets	8
Hosts per Subnet	30
Network Address	172.16.10.64
Broadcast Address	172.16.10.95
Host Range	172.16.10.65 – 172.16.10.94
172.16.10.78	Valid Host
172.16.10.95	Broadcast Address (Not Usable)

---

5. IPv6 Address

Given

2001:0db8:0000:0000:0000:ff00:0042:8329

Compressed Form

Remove leading zeros.

Replace consecutive zero blocks with ::

2001:db8::ff00:42:8329

Expanded Form
2001:0db8:0000:0000:0000:ff00:0042:8329

Network Prefix (/64)

First 64 bits

2001:0db8:0000:0000::/64

or

2001:db8:0:0::/64

Number of Host Addresses

Host bits

128−64=64128-64=64128−64=64

Total Hosts

2642^{64}264
=18,446,744,073,709,551,616=18,446,744,073,709,551,616=18,446,744,073,709,551,616

Approximately

18.4 Quintillion host addresses

Final Answer

Item	Result
Original Address	**2001:0db8:0000:0000:0000:ff00:0042:8329**
Compressed Address	**2001:db8::ff00:42:8329**
Expanded Address	**2001:0db8:0000:0000:0000:ff00:0042:8329**
Network Prefix (/64)	**2001:db8:0:0::/64**
Host Bits	**64**
Total Host Addresses	**2^{64} = 18,446,744,073,709,551,616**